

Week 12 - Global Energy: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: solar luminosity $L_{\odot} = 3.828 \times 10^{26}$ W, mean Earth–Sun distance $d = 1.496 \times 10^{11}$ m, Stefan–Boltzmann constant $\sigma_{\text{SB}} = 5.67 \times 10^{-8}$ W m⁻² K⁻⁴, Wien constant $b = 2.898 \times 10^{-3}$ m K.

YOU SHOULD TRY Question 1: From solar luminosity to the solar constant

Assume the Sun radiates isotropically.

1. Calculate the solar irradiance at the mean Earth–Sun distance using

$$S_0 = \frac{L_{\odot}}{4\pi d^2}.$$

2. Why is the global-mean incoming solar flux at the top of the atmosphere $S_0/4$ rather than S_0 ?
3. Calculate $S_0/4$.

IN CLASS Question 2: Earth's effective radiating temperature

Take Earth's planetary albedo to be $\alpha = 0.30$. In a simple zero-dimensional radiative equilibrium,

$$\frac{(1 - \alpha)S_0}{4} = \sigma_{\text{SB}}T_e^4.$$

1. Calculate the globally averaged absorbed solar flux.
2. Calculate T_e .
3. Compare it with a representative global-mean surface temperature of 288 K and calculate the difference.
4. As a simple grey diagnostic, write the outgoing longwave flux as

$$F_{\text{out}} = \epsilon_{\text{eff}}\sigma_{\text{SB}}T_s^4.$$

Using $T_s = 288$ K, calculate the effective emissivity ϵ_{eff} required for $F_{\text{out}} = F_{\text{abs}}$. Then substitute your value back into

$$T_s = \left(\frac{F_{\text{abs}}}{\epsilon_{\text{eff}}\sigma_{\text{SB}}} \right)^{1/4}$$

to verify that $\epsilon_{\text{eff}} \simeq 0.61$ corresponds to $T_s \simeq 288$ K. Is this the emissivity of the physical surface itself?

IN CLASS Question 3: Cloud droplet number, M_2 , extinction and albedo

Cloud A is $\Delta z = 500$ m deep and contains $N_A = 100 \text{ cm}^{-3}$ droplets of radius $r_A = 10 \text{ }\mu\text{m}$. Cloud B has the same liquid-water content and depth, but $N_B = 800 \text{ cm}^{-3}$. Assume identical droplets within each cloud.

For this simplified exercise define the area-related second moment as

$$M_2 = Nr^2,$$

and use the approximate cloud-albedo relation

$$A_c \approx \frac{X}{X + 15.4}, \quad X = \pi M_2 \Delta z.$$

Cloud-droplet extinction is approximately proportional to an area measure such as M_2 .

1. Use constant liquid-water content ($Nr^3 = \text{constant}$) to calculate r_B .
2. Convert the concentrations to m^{-3} and calculate M_2 for both clouds.
3. By what factor does M_2 change? What does this imply for extinction?
4. Calculate X and A_c for each cloud.
5. Why does doubling M_2 not double cloud albedo?
6. Show that, at fixed liquid-water content, $M_2 \propto N^{1/3}$. Hence estimate the fractional change in M_2 for a small 30% increase in N .
7. State the likely sign of the immediate shortwave effect on Earth's energy balance if a cloud becomes brighter in this way, all else equal.

YOU SHOULD TRY Question 4: Separate solar shortwave from terrestrial longwave

Use Wien's law to calculate the peak-emission wavelengths of

1. the Sun at 5778 K;
2. the Earth at 288 K.

Then identify the broad spectral region of each peak and state why this separation is important for the atmosphere.

IN CLASS Question 5: A one-layer greenhouse model

Consider a deliberately simple model atmosphere that is transparent to solar radiation but absorbs *all* terrestrial longwave radiation. The atmosphere emits as a blackbody equally upward and downward. Use the absorbed solar flux from Question 2, $F_{abs} = 238.2 \text{ W m}^{-2}$.

1. At the top of the atmosphere, radiative equilibrium requires $\sigma_{\text{SB}} T_a^4 = F_{abs}$. Calculate the atmospheric emitting temperature T_a .
2. The surface receives absorbed sunlight plus downward atmospheric longwave radiation. Show that its balance is

$$\sigma_{\text{SB}} T_s^4 = F_{abs} + \sigma_{\text{SB}} T_a^4 = 2F_{abs}.$$

Calculate T_s .

3. Compare this T_s with the effective radiating temperature from Question 2 and with the real mean surface temperature of about 288 K.
4. What does the fact that the one-layer model is too warm tell you about its assumptions?

EXTRA CHALLENGE Question 6: Venus: high albedo but a very hot surface

Venus is 1.08×10^{11} m from the Sun and has planetary albedo 0.70. For this question use solar luminosity 3.85×10^{26} W.

1. Calculate the solar irradiance at Venus.
2. Calculate its simple effective radiating temperature.
3. The actual Venusian surface temperature is about 730 K. Calculate how much hotter this is than the effective radiating temperature.
4. Explain why the surface can be so hot despite the high albedo.

EXTRA CHALLENGE Question 7: Use radiative imbalance to infer heat transport

At one latitude band, the annual-mean absorbed solar flux is 285 W m^{-2} and outgoing longwave radiation is 250 W m^{-2} . At a higher-latitude band, the corresponding values are 170 and 210 W m^{-2} .

1. Calculate the net top-of-atmosphere radiative imbalance in each band, taking positive as net energy gain.
2. Which band has an energy surplus and which a deficit?
3. In which direction must the climate system transport energy on average to prevent indefinite warming/cooling of the two bands?
4. Name two atmospheric/oceanic mechanisms that contribute to this transport.

YOU SHOULD TRY Question 8: Classify climate feedbacks from their causal chains

For each causal chain, classify the response as primarily *positive feedback*, *negative feedback*, or *sign depends on details*. Give a one-sentence explanation.

1. Warming $\rightarrow$ less snow/sea ice $\rightarrow$ lower albedo $\rightarrow$ more absorbed sunlight.
2. Warming $\rightarrow$ greater atmospheric water-vapour amount at similar RH $\rightarrow$ stronger infrared absorption.
3. Warming $\rightarrow$ larger thermal infrared emission to space.
4. Warming $\rightarrow$ a change in cloud amount, height, optical depth and phase.

EXTRA CHALLENGE Question 9: Match the Milankovitch cycle to its timescale and effect

Match each orbital variation to the most appropriate timescale from the list {26 000, 41 000, 100 000} years and state what aspect of Earth's orbit/orientation changes.

1. eccentricity;
2. obliquity (axial tilt);
3. precession.

Why can these cycles alter regional/seasonal climate even though they do not simply act like a permanent increase in the Sun's luminosity?